Fundamentals of Programming

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Books

- Turbo C 3 by Robert Lafore
- Deitel & Deitel :– C How to Program

Objectives

- Basics of Computer programming
- Programming Language

Basics of Computer Programming

- Computer Programming is the art of making a computer do what you want it to do.
- A computer program is simply a set of instructions to tell a computer how to solve a particular problem.

Basics of Computer Programming

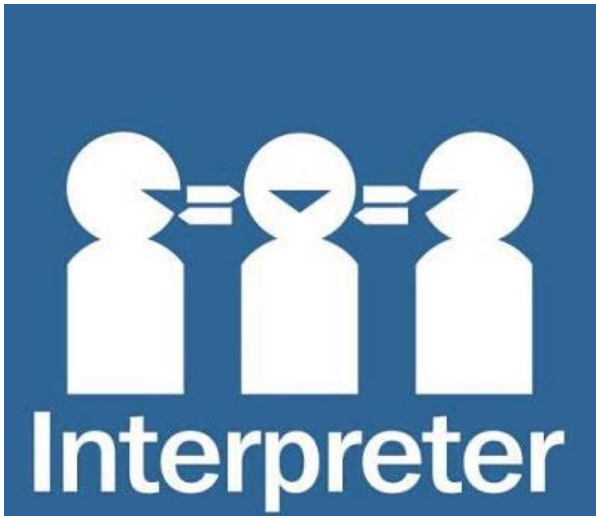
- It's rather like a recipe:
  - a set of instructions to tell a cook how to make a particular dish.
- It describes the ingredients (the data) and the sequence of steps (the process) needed to convert the ingredients into the cake or whatever.

Programming Language

- Just as you speak to a friend in a language so you 'speak' to the computer in a language. The only language that the computer understands is called *binary*.
- Binary is unfortunately very difficult for humans to read or write so we have to use an intermediate language and get it translated into binary for us.

Programming Language

- This is rather like watching the American and Russian presidents talking at a meeting.
  - One speaks in English, then an interpreter repeats what has been said in Russian. The other replies in Russian and the interpreter again repeats the sentence, this time in English.



Programming Language

- Surprisingly enough the thing that translates our intermediate language into binary is also called an interpreter.
- And just as you usually need a different interpreter to translate English into Russian than you do to translate Arabic into Russian so you need a different computer interpreter to translate Python into binary from the one that translates VBScript into binary.

Machine Code Programming

- The very first programmers actually had to enter the binary codes themselves, this is known as *machine code* programming and is incredibly difficult.

```
000010 00000 00000 00000 00100 000000   binary
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- The next stage was to create a translator that simply converted English equivalents of the binary codes into binary so that instead of having to remember that the code 001273 05 04 meant add 5 to 4 programmers could now write ADD 5 4.

Assembler Programming

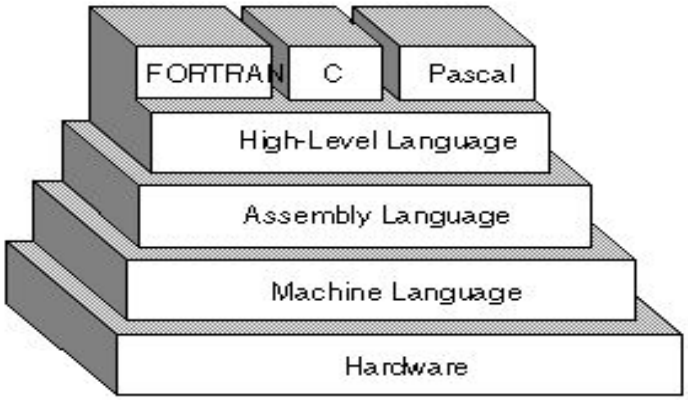
- This very simple improvement made life much simpler and these systems of codes were really the first programming languages, one for each type of computer. They were known as *assembler* languages and *Assembler programming* is still used for a few specialized programming tasks today.

Higher level computer languages

- It was still very difficult and took a lot of programming effort to achieve even simple tasks.
- Gradually computer scientists developed higher level computer languages to make the job easier. This was just as well because at the same time users were inventing ever more complex jobs for computers to solve!

Higher level computer languages

- This competition between the computer scientists and the users is still going on and new languages keep on appearing.



Points to remember

- Programs control the computer
- Programming languages allow us to 'speak' to the computer at a level that is closer to how humans think than how computers 'think'

Points to remember

- Basically a programmer writes a program in a high level language which is interpreted into binary that the computer understands.
- In technical speak the programmer generates *source code* and the interpreter generates *object code*. Sometimes object code has other names like: *binary code* or *machine code*.

Questions??